Lecture 2: Introduction to Fault Tolerance

Seyedhamidreza Mousavi

Outlines

- Fault tolerance
- Why do we need fault-tolerance?
- How to provide fault tolerance?
- History
- Applications
- Goal of fault tolerance

Fault tolerance

- Fault-tolerance is the ability of a system to continue performing its function in spite of faults
  - hardware   broken connection
  - software   bug in program

Why do we need fault-tolerance?

- It is practically impossible to build a perfect system
  - suppose a component has the reliability 99.99%
  - a system consisting of 100 non-redundant components will have the reliability 99.01%
  - a system consisting of 10,000 components will have the reliability 36.79%
- It is hard to foresee all the factors
  - unexpected environmental factor
  - potential user mistakes

How to provide fault tolerance

- to have fault tolerance in:
  - hardware component
    - Replicate it
  - a block of digital data
    - attach a parity check bit to it
  - the result of a software program
    - a line of program verifying its correctness
- The key is *REDUNDANCY*
- Redundancy is the provision of functional capabilities that would be unnecessary in a fault-free environment

History

- early computer systems
  - basic components had very low reliability
  - fault-tolerant techniques were need to overcome it
    - redundant structures with voting
    - error-detection and error correction codes

History

- advent of transistors
  - more reliable components
  - led to temporary decrease in the emphasis on fault-tolerant computing
  - designers thought it is enough to depend on the improved reliability of the transistor to guarantee correct computations

History

- last decades
  - more critical applications
    - space programs, military applications
    - control of nuclear power stations
    - banking transactions
  - VLSI made the implementation of many redundancy techniques practical and cost effective
  - Other than hardware component faults need to be tolerated:
    - transient faults (soft errors) caused by environmental factors
    - software faults

Applications

- safety-critical applications
  - critical to human safety
    - aircraft flight control
  - environmental disaster must be avoided
    - chemical plants, nuclear plants
  - requirements
    - 99.99999% probability to be operational at the end of a 3-hour period

Applications

- mission-critical applications
  - it is important to complete the mission
  - repair is impossible or prohibitively expensive
    - » Pioneer 10 was launched 2 March 1970, passed Pluto 13 June 1983
  - requirements
    - » 95% probability to be operational at the end of mission (e.g. 10 years)
    - » may be degraded or reconfigured before (operator interaction possible)

Applications

- business-critical applications
  - users want to have a high probability of receiving service when it is requested
  - transaction processing (banking, stock exchange or other time-shared systems)
    - » ATM: < 10 hours/year unavailable
    - » airline reservation: < 1 min/day unavailable

Goal of fault tolerance

The main goal of fault tolerance is

to increase the dependability of a system